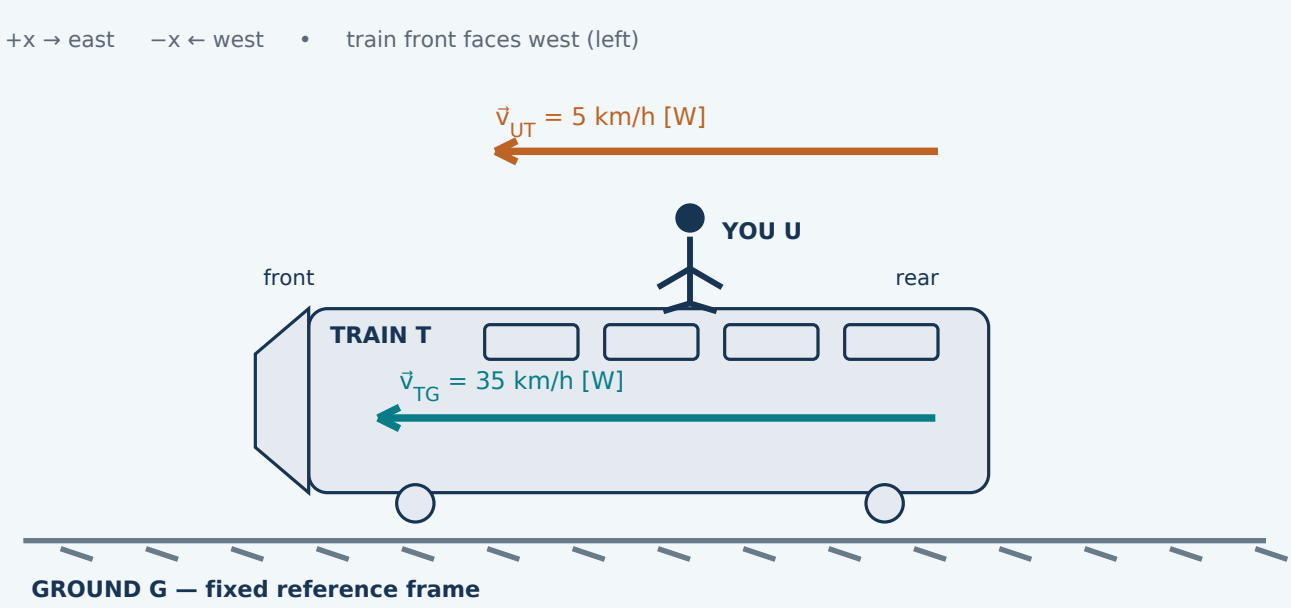
1.3a • Relative Motion

WORKED ANSWER KEY · SPH4U1

Notation: $\vec{v}_{AB}$ means A relative to B; signs follow the axes. **Precision:** Treat 250, 1200, and 30 min as two sig figs. Show intermediates to two decimals; calculate unrounded. A trailing decimal makes zero significant (10., 30., 40.).

1. Train moves 35 km/h [W]; you walk forward at 5 km/h. Which of five statements is wrong?



Step 1. Take the ground line as the fixed frame G. Choose east as +x, so the train's given westward velocity has a negative x component. T is the train and U is you.

$v_{TG,x} = -35$ km/h.

Step 2. You walk forward, which is also west. Your −5 km/h component is measured by someone on the train. A person on the ground sees that walking motion added to the train's motion.

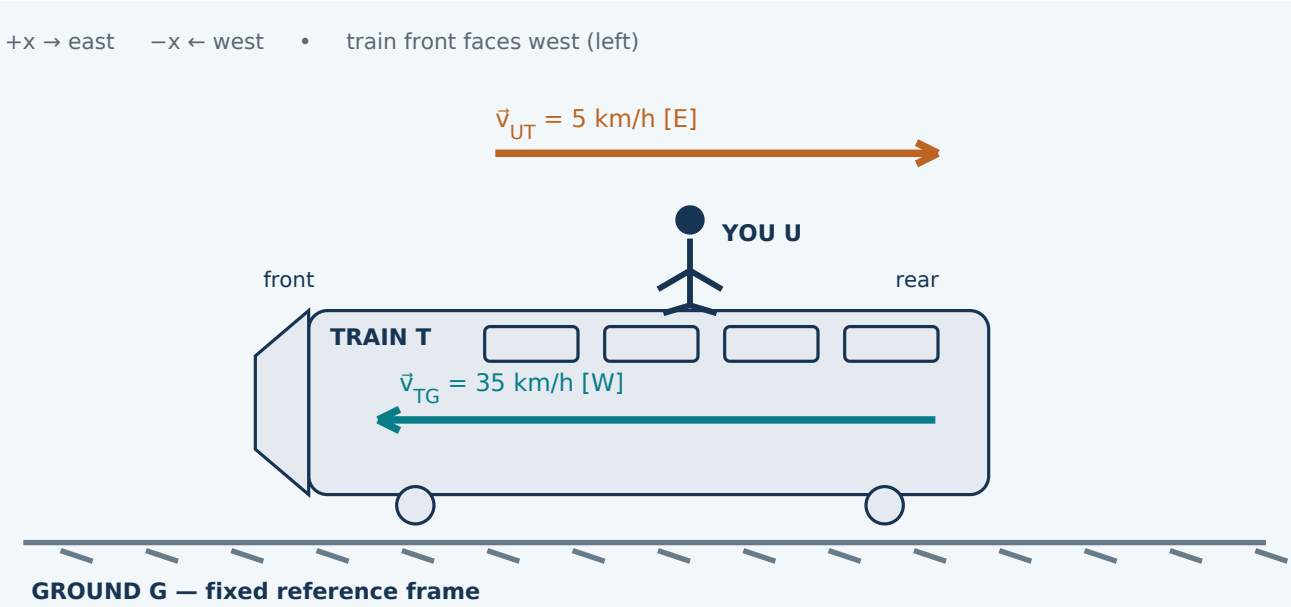
$v_{UT,x} = -5$ km/h; $v_{UG,x} = v_{TG,x} + v_{UT,x} = -35 + (-5) = -40$ km/h.

Step 3. Reverse object and observer to find the ground's velocity relative to you and to the train. Reversing changes the sign but keeps the same speed.

$v_{GU,x} = -v_{UG,x} = -(-40) = +40$ km/h; $v_{GT,x} = -v_{TG,x} = -(-35) = +35$ km/h.

Final answer: The fourth statement, "ground with respect to you is 35 km/h [E]," is wrong. It is 40. km/h [E] (to the ones place).

2. On the same westbound train you walk toward the rear at 5 km/h. Give the five velocities (a–e).



Step 1. The ground is fixed and east is +x. The train moves west (negative). The rear of the westbound train is east, so walking toward it is positive relative to the train.

$v_{TG,x} = -35$ km/h; $v_{UT,x} = +5$ km/h.

Step 2. Add your walking component to the train's ground component. The answer is still negative: you move west over the ground, just 5 km/h more slowly than the train.

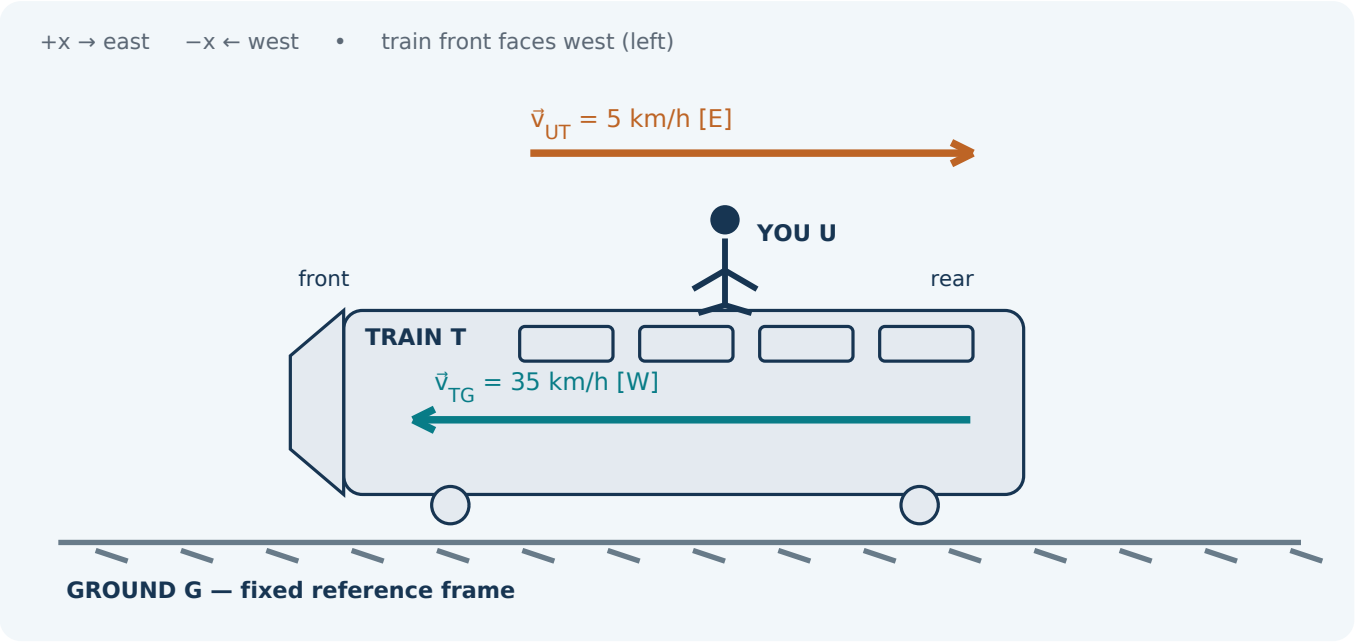
$v_{UG,x} = v_{TG,x} + v_{UT,x} = -35 + 5 = -30$ km/h.

Step 3. For the ground relative to you, reverse your ground velocity. For the ground relative to the train, reverse the train's ground velocity.

$v_{GU,x} = -(-30) = +30$ km/h; $v_{GT,x} = -(-35) = +35$ km/h.

Final answer: (a) Train relative to ground: 35 km/h [W].
(b) You relative to train: 5 km/h [E].
(c) You relative to ground: 30. km/h [W].
(d) Ground relative to you: 30. km/h [E].
(e) Ground relative to train: 35 km/h [E].

4. Rewrite the five results of Question 2 using object/frame subscripts.



Step 1. The first subscript names what moves; the second names the observer. For example, $\vec{v}_{UT}$ is your velocity as seen from the train. East is positive in the calculations.

$$v_{TG,x} = -35 \text{ km/h}; \; v_{UT,x} = +5 \text{ km/h}.$$

Step 2. (a) and (b) are the two given vectors. For (c), combine them to express your motion in the ground frame.

$$v_{UG,x} = v_{TG,x} + v_{UT,x} = -35 + 5 = -30 \text{ km/h}.$$

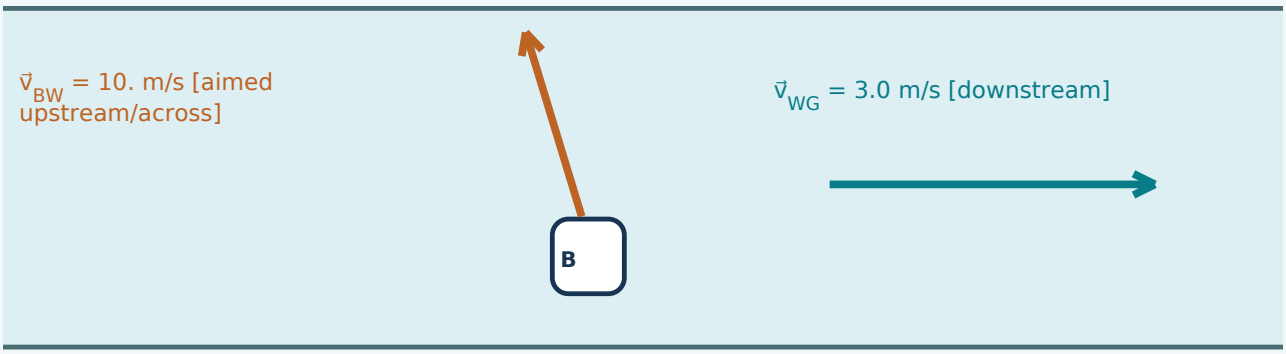
Step 3. (d) Ground relative to you reverses (c). (e) Ground relative to train reverses (a). In each pair, object and observer swap places, so the velocity reverses.

$$v_{GU,x} = -v_{UG,x} = +30 \text{ km/h}; \; v_{GT,x} = -v_{TG,x} = +35 \text{ km/h}.$$

Final answer: (a) $\vec{v}_{TG}$ = 35 km/h [W].
(b) $\vec{v}_{UT}$ = 5 km/h [E].
(c) $\vec{v}_{UG}$ = 30. km/h [W].
(d) $\vec{v}_{GU}$ = 30. km/h [E].
(e) $\vec{v}_{GT}$ = 35 km/h [E].

River a–e. Boat speed through water is 10. m/s; river current is 3.0 m/s downstream. Compare aiming upstream to aiming straight across and identify a, b, c.

FIG. 1 AIM UPSTREAM



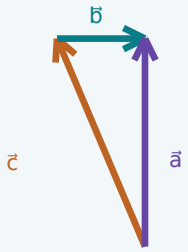
SHORE / GROUND G: fixed reference

+x → downstream −x ← upstream +y ↑ across −y ↓ back

Step 1. The shore is the fixed frame G. The current is water relative to shore, $\vec{v}_{WG}$; the aiming arrow is boat relative to water, $\vec{v}_{BW}$. Add them to find boat relative to shore, $\vec{v}_{BG}$. On the worksheet these are $\vec{b}$, $\vec{c}$, and $\vec{a}$.

$\vec{v}_{WG}$ = (+3.0, 0) m/s; $\vec{v}_{BG} = \vec{v}_{WG} + \vec{v}_{BW}$; $\vec{a} = \vec{b} + \vec{c}$.

VELOCITY TRIANGLE • vectors joined head to tail



- $\vec{c} = \vec{v}_{BW}$ 10. m/s [upstream/across]
- $\vec{b} = \vec{v}_{WG}$ 3.0 m/s [downstream]
- $\vec{a} = \vec{v}_{BG}$ 9.54 m/s [across]

Diagram shows velocity addition; arrow endpoints are not positions on a map.

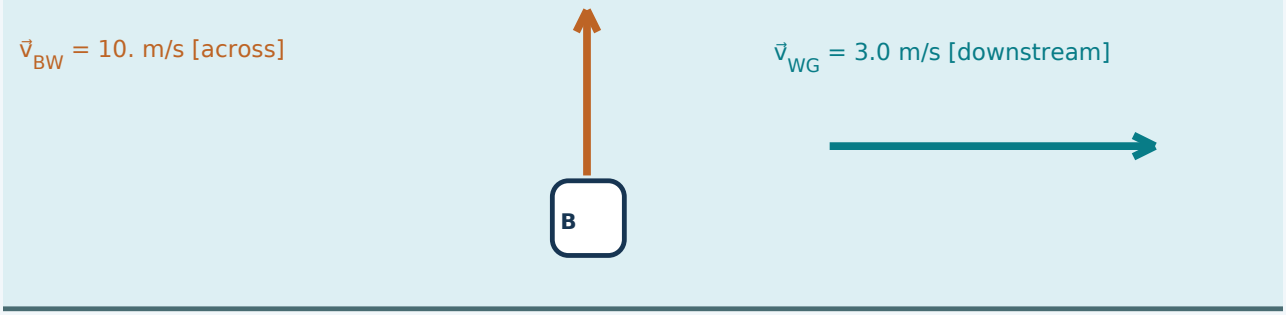
Step 2. Figure 1: aim partly upstream. The boat needs a −3.0 m/s downstream component to cancel the +3.0 m/s current. Use the 10. m/s boat speed to find its across component.

$v_{BW,across} = \sqrt{[(10.)^2 - (3.0)^2]} \approx 9.54$ m/s; $\theta_{aim} = \sin^{-1}(3.0/10.) \approx 17.46°$.

$\vec{v}_{BG,1}$ = (−3.0, $+\sqrt{[(10.)^2 - (3.0)^2]}$) + (+3.0, 0) ≈ (0, +9.54) m/s.

River a–e • continued: straight-across aim

FIG. 2 AIM STRAIGHT ACROSS



SHORE / GROUND G: fixed reference

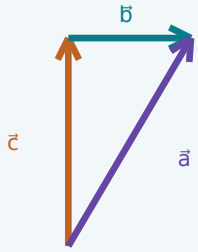
+x → downstream −x ← upstream +y ↑ across −y ↓ back

Step 3. Figure 2: aim straight across relative to the water. The boat has an across component of 10. m/s and the current adds 3.0 m/s downstream. Use the unrounded expression to determine the final speed.

$\vec{v}_{BG,2} = (0, +10.) + (+3.0, 0) = (+3.0, +10.)$ m/s.

$|\vec{v}_{BG,2}| = \sqrt{(3.0^2 + 10.^2)} \approx 10.44$ m/s; $\theta = \tan^{-1}(3.0/10.) \approx 16.70°$.

VELOCITY TRIANGLE • vectors joined head to tail



- $\vec{c} = \vec{v}_{BW}$ 10. m/s [across]
- $\vec{b} = \vec{v}_{WG}$ 3.0 m/s [downstream]
- $\vec{a} = \vec{v}_{BG}$ 10.44 m/s [17° downstream]

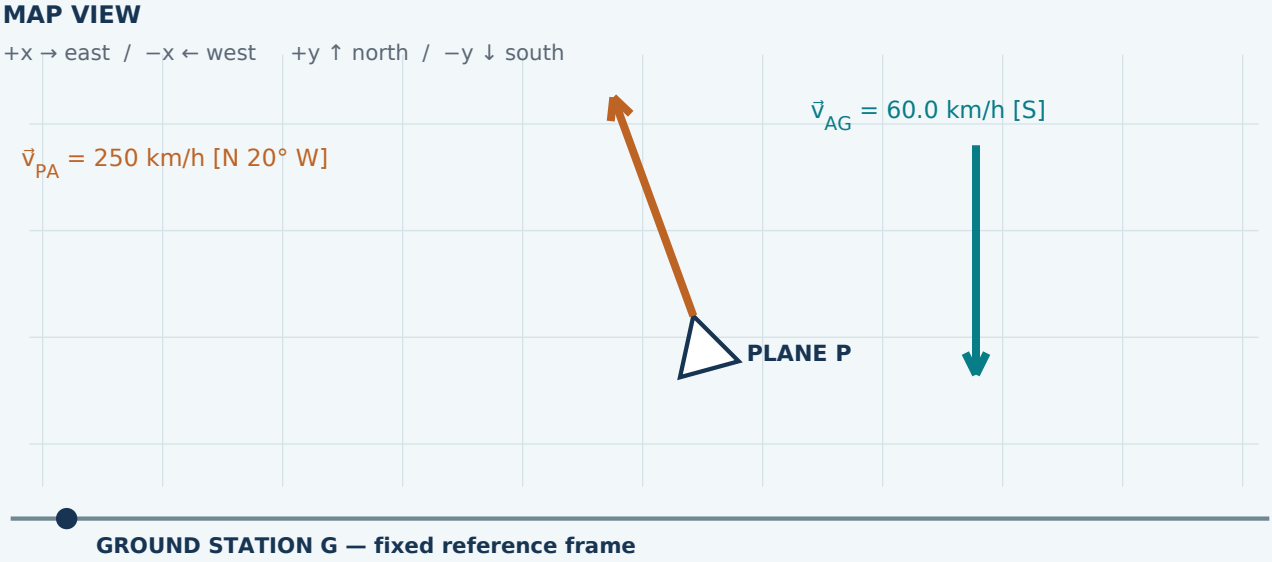
Diagram shows velocity addition; arrow endpoints are not positions on a map.

Step 4. If the river width is D, crossing time depends on ground velocity across the river. Figure 2 uses the full 10. m/s across, so it takes less time.

$t_1 = D/\sqrt{[(10.)^2 - (3.0)^2]} \approx D/(9.54 \text{ m/s})$; $t_2 = D/(10. \text{ m/s})$; $t_2 < t_1$.

Final answer: (a) Figure 2 crosses first.
(b) $\vec{c} + \vec{b} = \vec{a}$.
(c–d) $\vec{c} = \vec{v}_{BW}$ (boat relative to water, 10. m/s); $\vec{b} = \vec{v}_{WG}$ (current, 3.0 m/s downstream); $\vec{a} = \vec{v}_{BG}$ (boat relative to shore).
(e) Figure 1: 9.5 m/s [across]. Figure 2: 10. m/s [17° downstream of across].

Application 1. Plane heads [N 20° W] at 250 km/h in a north wind of 60.0 km/h for 30 min. Find its ground velocity and displacement.



Step 1. Use the ground station G as the reference. The plane's heading is relative to the air; a north wind blows from north, so the air moves south. Join the two given arrows head to tail to get the plane's ground velocity.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}.$$



Diagram shows velocity addition; arrow endpoints are not positions on a map.

Step 2. Resolve the 250 km/h heading into a westward x component and a northward y component. The 20° angle is measured from north, so x uses sine and y uses cosine.

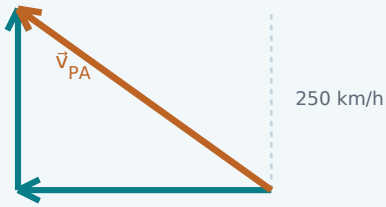
$$v_{PA,x} = -250 \sin 20° \approx -85.51 \text{ km/h}.$$

$$v_{PA,y} = +250 \cos 20° \approx +234.92 \text{ km/h}.$$

Application 1 • continued: calculate the resultant

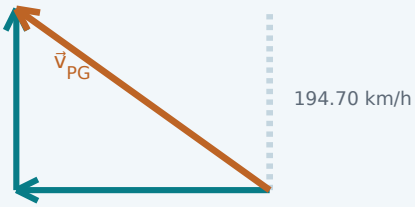
COMPONENT TRIANGLES • +x east, +y north

Plane through air [N 20° W]



$v_{PA,x} = -85.51$ km/h
$v_{PA,y} = +234.92$ km/h

Plane over ground [W 63.95° N]



$v_{PG,x} = -85.51$ km/h
$v_{PG,y} = +174.92$ km/h

The signed components go into the calculation; the diagonal is the vector magnitude and direction.

Step 3. The wind has no east–west component and a negative north–south component. Add it to the plane components: it reduces the northward motion but does not change the westward motion.

$v_{AG,x} = 0$; $v_{AG,y} = -60.0$ km/h.

$v_{PG,x} = -250 \sin 20° \approx -85.51$ km/h; $v_{PG,y} = 250 \cos 20° - 60.0 \approx +174.92$ km/h.

Step 4. The ground components make a right triangle. Apply Pythagoras to their magnitudes, using the unrounded trig expressions in the calculator.

$|\vec{v}_{PG}| = \sqrt{[(250 \sin 20°)^2 + (250 \cos 20° - 60.0)^2]} \approx 194.70$ km/h.

Step 5. For the bearing in your [W θ N] form, the north component is opposite θ and the west magnitude is adjacent. [W 63.95° N] is the same direction as [N 26.05° W].

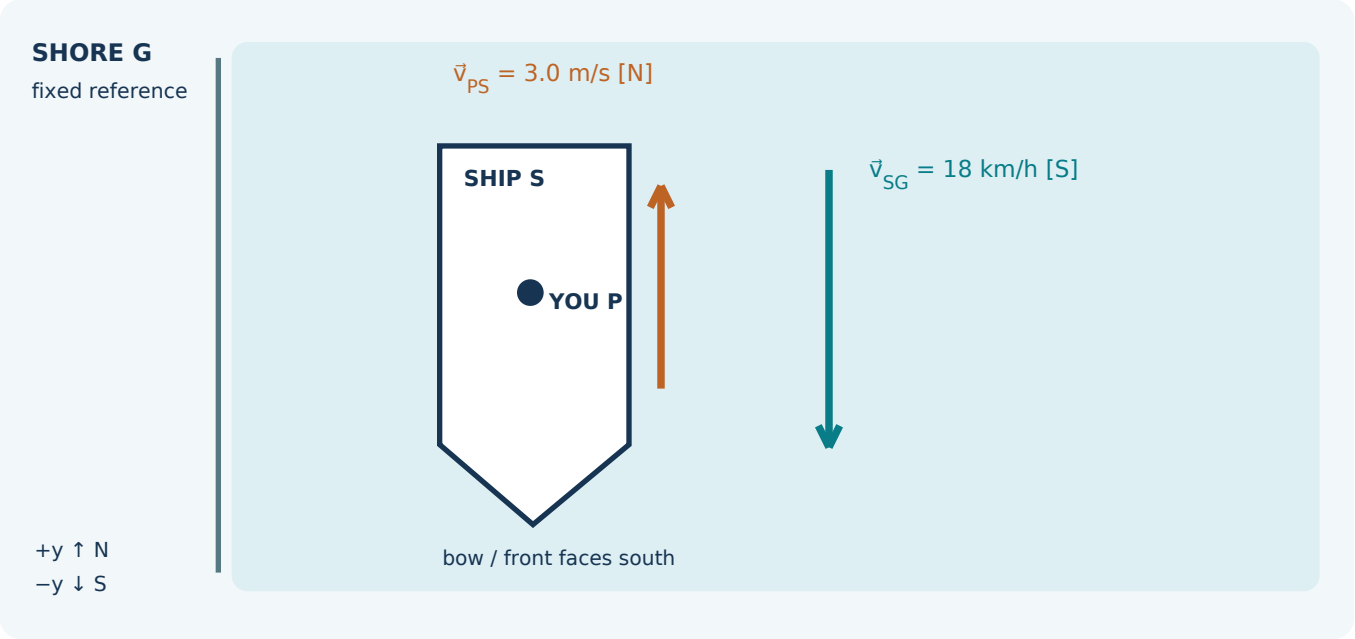
$\theta = \tan^{-1}[(250 \cos 20° - 60.0)/(250 \sin 20°)] \approx 63.95°$ north of west.

Step 6. Convert 30 min to 0.50 h. Displacement has the same direction as the constant ground velocity; its magnitude is ground speed × time.

$|\Delta\vec{r}_{PG}| = (0.50 \text{ h})\sqrt{[(250 \sin 20°)^2 + (250 \cos 20° - 60.0)^2]} \approx 97.35$ km.

Final answer: Ground velocity: 1.9×10^2 km/h [W 64° N] (also [N 26° W]).
Displacement: 97 km [W 64° N].

Application 2. Ship travels 18 km/h south; passenger walks toward the rear at 3.0 m/s. Find passenger velocity over ground.



Step 1. The shore line is fixed ground G. Choose north +y. Convert the ship's given speed to m/s before adding it to the walking velocity. South makes the ship's component negative.

$18\ \text{km/h} \times (1000\ \text{m/km})/(3600\ \text{s/h}) = 5.0\ \text{m/s};\ v_{SG,y} = -5.0\ \text{m/s}.$

Step 2. The rear of a southbound ship is north. The passenger walks north relative to the ship, so that component is positive.

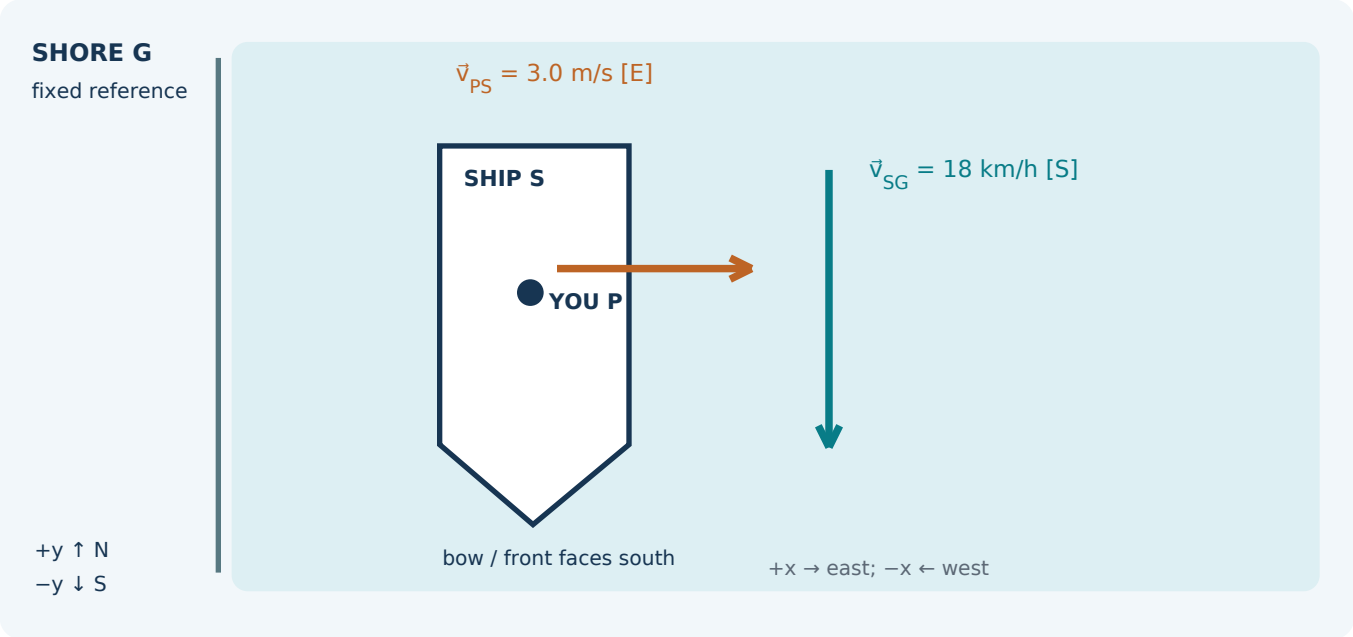
$v_{PS,y} = +3.0\ \text{m/s}.$

Step 3. Add the ship's velocity over the ground to the passenger's walking velocity on the ship. The negative result points south.

$v_{PG,y} = v_{SG,y} + v_{PS,y} = -5.0 + 3.0 = -2.0\ \text{m/s}.$

Final answer: Passenger relative to ground: 2.0 m/s [S] (7.2 km/h [S]).

Application 3. The same passenger turns right from walking toward the rear and walks at 3.0 m/s toward the railing. Find velocity over ground.



Step 1. The ship still moves south at 18 km/h. Convert once so both given speeds have the same units. The shore G is fixed; choose east +x, north +y.

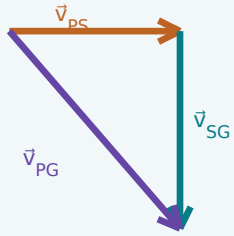
$\vec{v}_{SG} = (0, -5.0)$ m/s.

Step 2. The passenger was facing the rear (north). Turning right makes the new walking direction east relative to the ship.

$\vec{v}_{PS} = (+3.0, 0)$ m/s.

Application 3 • continued: add the velocities

VELOCITY TRIANGLE • vectors joined head to tail



- $\vec{v}_{PS}$ 3.0 m/s [E]
- $\vec{v}_{SG}$ 5.0 m/s [S]
- $\vec{v}_{PG}$ 5.83 m/s [E 59.04° S]

Diagram shows velocity addition; arrow endpoints are not positions on a map.

Step 3. Join the east walking arrow to the south ship arrow. Their resultant is the passenger's velocity seen from shore.

$$\vec{v}_{PG} = \vec{v}_{PS} + \vec{v}_{SG} = (+3.0, -5.0) \text{ m/s.}$$

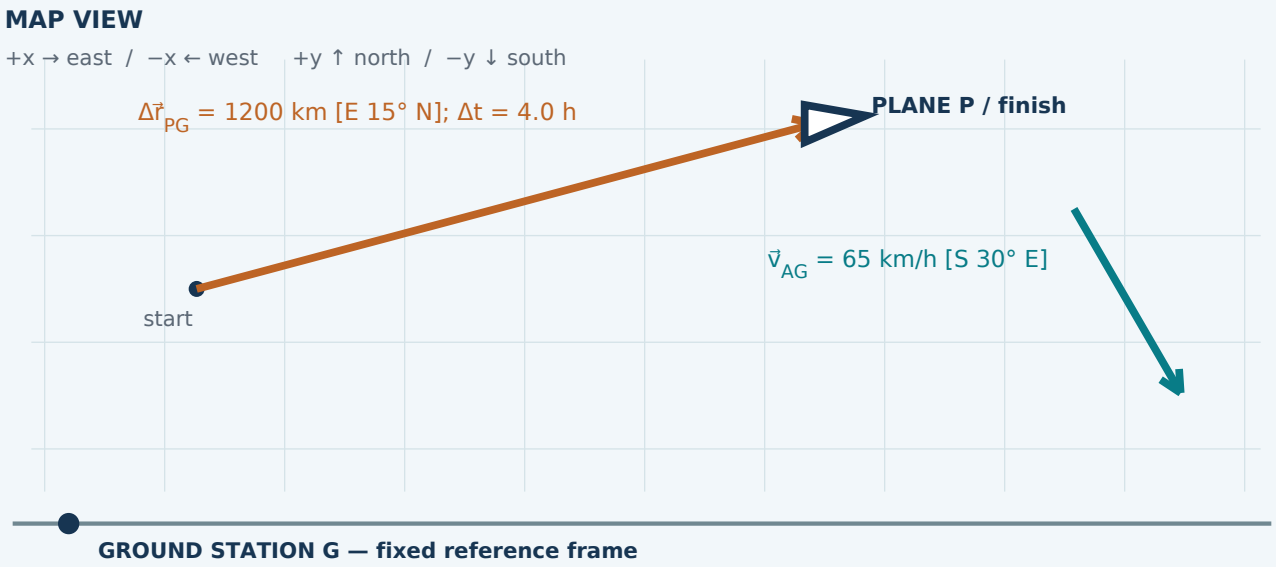
Step 4. Use the right triangle to find the ground speed. Measure the angle from east down toward south, as in your sketch.

$$|\vec{v}_{PG}| = \sqrt{(3.0^2 + 5.0^2)} \approx 5.83 \text{ m/s.}$$

$$\theta = \tan^{-1}(5.0/3.0) \approx 59.04° \text{ south of east.}$$

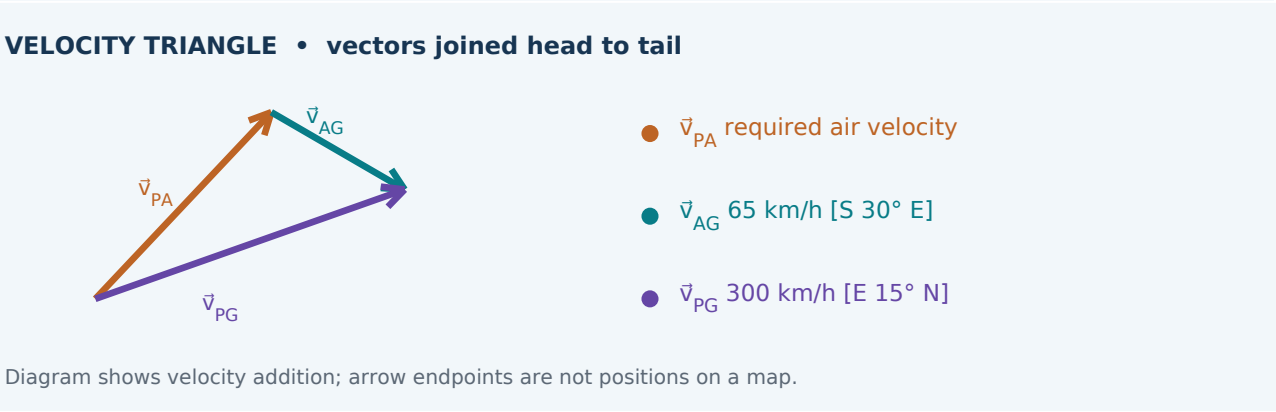
Final answer: Passenger relative to ground: 5.8 m/s [E 59° S] (also [S 31° E]).

Application 4. Wind is 65 km/h [S 30° E]; plane travels 1200 km [E 15° N] in 4.0 h. Find required air velocity.



Step 1. The ground station G is fixed. The given route is a ground displacement [E 15° N], so divide its magnitude by the elapsed time to get the ground speed.

$|\vec{v}_{PG}|$ = 1200 km / 4.0 h = 300 km/h.



Step 2. Resolve the 300 km/h ground velocity. The route angle is 15° above east, so x uses cosine and y uses sine.

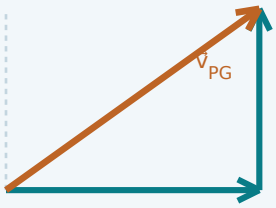
$v_{PG,x}$ = +300 cos15° ≈ +289.78 km/h.

$v_{PG,y}$ = +300 sin15° ≈ +77.65 km/h.

Application 4 • continued: calculate the resultant

COMPONENT TRIANGLES • +x east, +y north

Ground route [E 15° N]

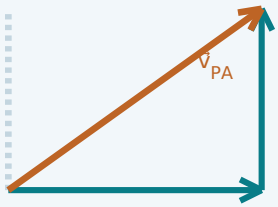


300 km/h

$v_{PG,x} = +289.78$ km/h

$v_{PG,y} = +77.65$ km/h

Required air heading [E 27.50° N]



290.05 km/h

$v_{PA,x} = +257.28$ km/h

$v_{PA,y} = +133.94$ km/h

The signed components go into the calculation; the diagonal is the vector magnitude and direction.

Step 3. Resolve the wind [S 30° E]. It moves east and south; south is negative on our y axis.

$v_{AG,x} = +65 \sin 30° = +32.50$ km/h.

$v_{AG,y} = -65 \cos 30° \approx -56.29$ km/h.

Step 4. Ground velocity is air velocity plus wind. The ground route is known, so subtract the wind vector to find the required velocity through air.

$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG} \Rightarrow \vec{v}_{PA} = \vec{v}_{PG} - \vec{v}_{AG}$.

Step 5. Subtract each signed component. Subtracting the southward wind component adds northward speed.

$v_{PA,x} = 300 \cos 15° - 65 \sin 30° \approx +257.28$ km/h.

$v_{PA,y} = 300 \sin 15° - (-65 \cos 30°) \approx +133.94$ km/h.

Step 6. Use the component triangle for airspeed. Keep the exact trig expressions in the calculator so rounding the displayed components does not change the result.

$|\vec{v}_{PA}| = \sqrt{[(300 \cos 15° - 65 \sin 30°)^2 + (300 \sin 15° + 65 \cos 30°)^2]} \approx 290.05$ km/h.

Step 7. Both components are positive, so the heading is north of east. The north component is opposite the angle and the east component is adjacent.

$\theta = \tan^{-1}[(300 \sin 15° + 65 \cos 30°)/(300 \cos 15° - 65 \sin 30°)] \approx 27.50°$.

Final answer: Required airspeed and heading: 2.9×10^2 km/h [E 28° N].